



The Antidiabetic Metformin as an Adjunct to Antidepressants in Patients with Major Depressive Disorder: A Proof-of-Concept, Randomized, Double-Blind, Placebo-Controlled Trial

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Abstract

Metformin (MET) has been reported to have antidepressant effects in animal models and nondiabetic patients with depression, owing to its anti-inflammatory, antioxidant, and neuroprotective activity. Accordingly, we proposed that MET would show antidepressant effects in patients with major depressive disorder (MDD) without other comorbidities. In this double-blind placebo-controlled study, 80 adult outpatients with MDD (DSM-IV criteria) and a Hamilton Depression Rating Scale (HAM-D) score >18 were randomized to receive fluoxetine 20 mg once daily plus placebo (n = 40) or fluoxetine 20 mg once daily plus MET 1000 mg once daily for 12 weeks. Patients were assessed by HAM-D score (weeks 0, 4, 8, and 12). The serum levels of TNF- α , IL-1 β , IL-6, IGF-1, MDA, CRP, BDNF, and serotonin were measured before and after therapy. Mixed-effects model repeated-measures analysis of covariance was used to compare the HAM-D scores and the biological markers between the two groups. After 4, 8 and 12 weeks, patients in the MET group showed a statistically significant decline in HAM-D score relative to the placebo group (least squares mean difference [LSMD] -2.34 , $p = 0.000$, LSMD -3.369 , $p = 0.000$, and LSMD -3.454 , $p = 0.000$, respectively). Response and remission rates were significantly higher in the MET group (89% and 81%, respectively) than in the placebo group (59% and 46%, respectively). Moreover, the MET group was superior in conserving the measured biological markers compared with the placebo group. Our findings suggest MET as a promising, effective, and safe short-term adjunctive approach in nondiabetic MDD patients. **Trial registration ID: NCT04088448.**

Key Words Major depressive disorder · Metformin · Fluoxetine · Inflammatory markers · BDNF · Adjunctive therapy

Introduction

Major depressive disorder (MDD) remains largely refractive to current therapeutic approaches, which are restricted to the regulation of monoamine transmission modulation [1]. In recent decades, various strategies for MDD treatment have been

developed to improve response and remission rates [2]. Recent evidence indicates a correlation between depression and inflammatory factors within the innate and adaptive immune systems [3]. Consequently, the implementation of safe, new adjunctive treatment for MDD is urgently needed to overcome resistance and boost the therapeutic response [4].

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Expanding evidence shows that inflammation may play a crucial role in MDD pathophysiology [5]; the release of pro-inflammatory cytokines regulates monoamine metabolism [6]. Furthermore, inflammatory cytokines may influence astrocytes, leading to a reduction in glutamate reuptake and increase in its release, together with a decrease in the synthesis of brain-derived neurotrophic factor (BDNF), which has an impact on neuronal integrity and neurogenesis [7]. Recent clinical studies have demonstrated that patients with MDD have elevated serum levels of pro-inflammatory cytokines, including tumor necrosis factor alpha (TNF- α) and interleukins IL-1b and IL-6 [8, 9]. The results of these studies showed an improvement in mood and enhanced antidepressant response as a result of the suppression of cytokine signaling in MDD patients [10].

Several clinical studies have suggested that anti-inflammatory agents, administered either as monotherapy or in addition to antidepressants, may exert antidepressant effects in patients with depressive episodes [11–13]. Some antidepressant drugs were also found to elicit anti-inflammatory and neuroprotective effects, partly due to their influence on cytokine production [14, 15].

Insulin-like growth factor 1 (IGF-1) is a neurotrophic/growth factor that has also been found to be involved in antidepressant response [16]. IGF-1 develops in the brain, and has an effect on mood control [17]. It was reported that IGF-1 was significantly higher in depressed patients relative to healthy controls [18].

Metformin (MET) is commonly used as a first-line therapy for patients with type 2 diabetes mellitus to minimize hepatic glucose output and improve the insulin-mediated uptake of glucose [19]. MET has the ability to reduce the adhesion of inflammatory cells to the endothelium; it also has neuroprotective, anti-inflammatory, antiapoptotic, and antioxidant properties [20, 21]. MET has been shown to enhance antidepressant efficacy and improve cognition in preclinical studies [22, 23]. It has also been reported that MET may enhance the recovery of depression comorbid with type 2 diabetes mellitus by improving cognitive performance [24, 25]. These meaningful outcomes suggest it to be an attractive candidate as an adjuvant therapy for MDD.

In this trial, we hypothesized that MET would show an antidepressant effect in depressed patients without other comorbidities. In the present double-blind placebo-controlled study, we aimed to evaluate the adjunctive effect of MET with fluoxetine in the treatment of patients with MDD who did not have other problems. We also assessed the relationship between the Hamilton Depression Rating Scale (HAM-D) score and several peripheral biomarkers and their role in diagnosing MDD and its therapeutic outcomes.

Materials and Methods

Study Design

This was a multicenter, prospective, randomized, double-blinded, placebo-controlled study, which was conducted in both Abou El Azayem Psychiatric Hospital in 10th of Ramadan and Menoufia University Hospital, Egypt (January 2017 to December 2019).

Participants

Eligible patients were individuals aged 23–65 years with a diagnosis of MDD based on the Diagnostic and Statistical Manual of Mental Disorders IV (DSM-IV) Mini-International Neuropsychiatric Interview (MINI) [26, 27], and HAM-D score >18, with item 1 (depressed mood) scored 2 or greater [28]. All patients and their legally authorized representatives provided informed consent in accordance with the procedures outlined by the local ethical committees, and were informed that they could withdraw from the trial at any time. The protocol was approved by the ethical committees of both the Abou El Azayem Psychiatric Hospital and the Faculty of Medicine, Menoufia University, Egypt. The study was performed in accordance with the ethical standards as laid down in the 1964 Declaration of Helsinki and its later amendments or comparable ethical standards.

Patients with bipolar disease, seasonal depression, personality disorders, eating disorders, drug dependency or abuse, concurrent active medical condition or history of seizures, inflammatory disorders, or drug allergy or contraindications were excluded from the study. In addition, all patients who had taken other psychotropic agents, including antidepressants, within the prior 4 weeks or had undergone electroconvulsive therapy within the prior 2 months were excluded. Pregnant or lactating women, patients with serious disease, those who fulfilled the metabolic syndrome criteria, and patients with diabetes, liver disease, or heart failure were also not eligible for the study. All the screened patients were thoroughly tested for the occurrence of diabetes, metabolic syndrome, and liver or heart disease in a comprehensive clinical, electrocardiographic, and laboratory examination.

Demographic Data

Patients' medical history was taken to ensure the absence of any interacting or interfering drugs and diseases. Demographic data were collected at baseline using a structured questionnaire. The questionnaire included age, gender, weight, height, body mass index (BMI), marital status, episodes of depression, drugs used in last episode, and familial history of MDD.

Sample Size

Calculation of sample size based on a meta-analysis of antidepressant treatment trials showed that placebo treatment has an average effect size of 1.69 compared with 2.50 for an antidepressant treatment [29]. Using an 80% power and two-sided significance of 5% with an effect size of 0.81, the sample size was 26 subjects per group. A final sample size of 30 subjects was estimated, assuming a 15% attrition rate. Therefore, our sample size, 40 per group, should have adequate power to test our hypothesis.

Randomization and Blinding

Using a computerized random number generator, we randomized study participants in a 1:1 ratio into blocks of four to receive either MET or placebo in addition to their standard treatment, according to the Consolidated Standards of Reporting Trials (CONSORT) guidelines. Allocation concealment was achieved using numbered opaque envelopes which were sequentially scanned and stapled. Randomization and allocation, as well as interviews, were performed by different individuals. The physician who referred the patient, the patients themselves, the resident who administered the drugs and rated the patients, and the statistician were all blinded to the allocation. The responsible psychiatrist was unblinded only if the patient's trial drug had an effect on immediate emergency therapy. Once the blinding had been broken, the patient was managed as off-trial. Participants were withdrawn from the study if they missed seven consecutive days of the trial medication.

Intervention

Forty patients in the placebo group received fluoxetine 20 mg once daily plus one placebo tablet, while another 40 patients in the MET group received fluoxetine 20 mg once daily plus 1000 mg XR MET tablet once daily with food for 12 weeks. Placebo tablets were dispensed by Sigma Pharmaceutical Industries, Menoufia, Egypt. Trial medications were dispensed by the trial pharmacist and the returned medications were audited.

Outcome

The main outcome of the study was the 17-item HAM-D score, which was measured at baseline and after 4, 8, and 12 weeks from the start of therapy. Remission was defined as a HAM-D total score ≤ 7 (primary outcome). Treatment response was defined as $\geq 50\%$ drop in the HAM-D total score (secondary outcome).

The patients' conditions were monitored to check adherence to and possible side effects of the medications via questioning using a checklist. The patients were followed up weekly by phone for assessment of compliance with the study medication, adverse events, and any signs of infection or

inflammation. The tablets remaining in each supply given to the patients were counted to evaluate treatment compliance.

Serum levels of TNF- α , IL-1 β , IL-6, IGF-1, malondialdehyde (MDA), high-sensitivity C-reactive protein (hsCRP), BDNF, and serotonin were measured at baseline and after therapy to evaluate the biological effects of the study medications.

Measurements

Blood samples were collected for all patients at the same time point for everyone, with a fasting morning sample by venipuncture into plain vacutainers. The tubes were then centrifuged at 4500 $\times g$ for 10 min. The separated samples were transferred to Eppendorf tubes and kept in a freezer at -80°C until analysis. The serum levels of TNF- α , IL-1 β , IL-6, IGF-1, MDA, CRP, BDNF, serotonin, and vitamin B12 were measured with specific commercial enzyme-linked immunosorbent assay (ELISA) kits, which were purchased from MyBioSource, Inc. (USA). All measurements were performed according to the manufacturer's specifications using a Biotek ELx800 UV-Visible plate reader (USA).

Statistical Analysis

We report continuous variables as mean $\pm$ standard deviation (SD) and categorical variables as number (percentage) unless stated otherwise. All tests of treatment efficacy were conducted using a two-sided significance level of 0.05, and Bonferroni adjustments were made for multiple comparisons. Type III sums of squares were used to adjust unbalanced data in the interactions of these models of variance. Mixed-effects model repeated-measures (MMRM) analysis of covariance (ANCOVA) was used for the primary analysis of any change from baseline to endpoint in HAM-D total score.

In addition, two-way repeated-measures analysis of variance (ANOVA) was performed for HAM-D scores (time-treatment interaction). The two groups as a between-subject factor (group) and the four interval measurements during treatment as the within-subject factor (time) were considered. ANCOVA, controlling for the baseline score, was used to compare the change in biological markers at the 12th week between the two groups. Fisher's exact test was used for the qualitative variables. Pearson's correlation was calculated to assess the relationship among variables. The statistical analysis was performed using IBM® SPSS® Statistics version 22 software (IBM Corp., Armonk, NY, USA). All graphs were created with GraphPad Prism 6.01 software (GraphPad Software, La Jolla CA, USA).

Results

One hundred and twenty patients were screened for the study. Forty patients were excluded from the study because they had

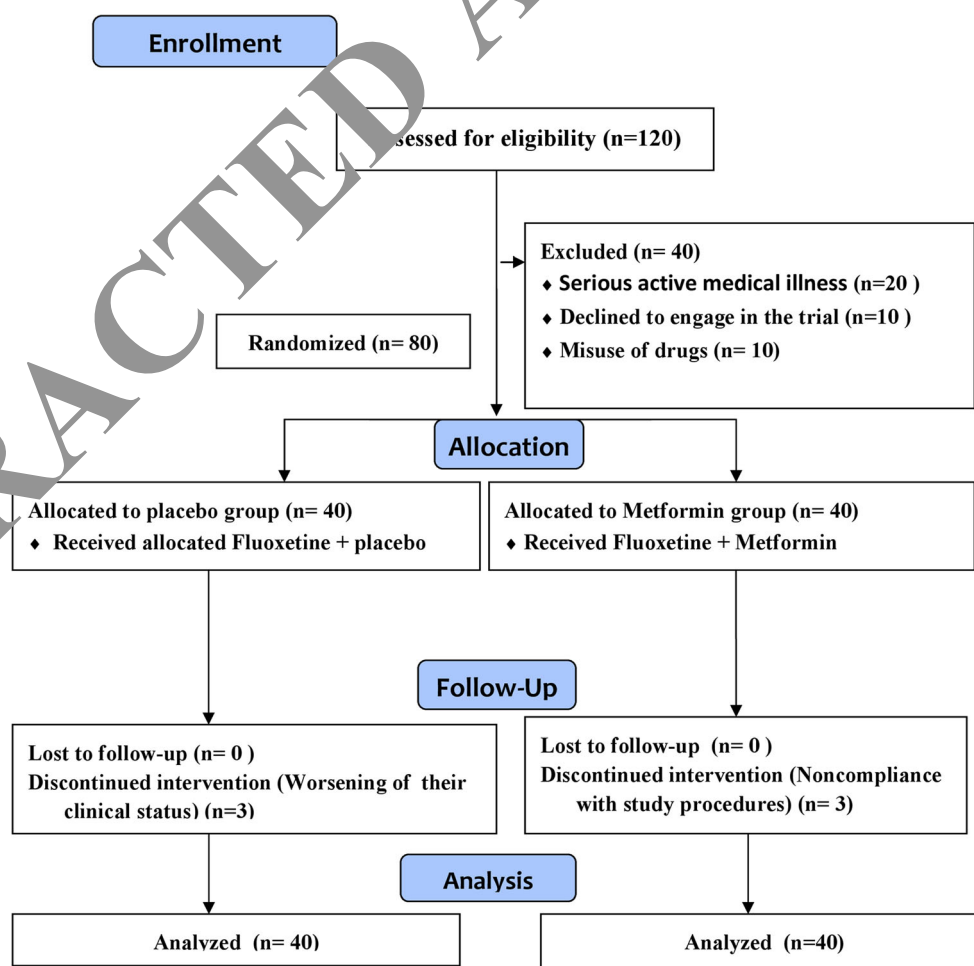
other serious active medical illness, misuse of drugs, or declined to engage in the trial. Ultimately, 80 patients were recruited and randomized to the trial, as shown in Fig. 1.

There were no statistically significant differences between patients assigned to the placebo and MET groups regarding their demographic data (Table 1). Six patients dropped out 4 weeks after commencement of the trial: three from the placebo group who experienced worsening of their clinical status, and the other three from the MET group due to noncompliance with study procedures. These six subjects were included in the HAM-D analysis using MMRM ANCOVA, but they were excluded from the biological marker analysis, as only the baseline data were available.

Effect on HAM-D Score (Primary Outcome)

No statistically significant difference in HAM-D score was found between the placebo and MET groups at baseline ($p > 0.05$). The response rate was 89% for the MET group vs. 59% for the placebo group ($p = 0.000$; number needed to treat [NNT] = 4). The remission rate was 81% for the MET group vs. 46% for the placebo group ($p < 0.013$; NNT = 3.33).

Fig. 1 Flow diagram of study participants



The MET group showed a statistically significant greater improvement in the HAM-D total score than the placebo group after 4, 8, and 12 weeks from the start of treatment using the primary MMRM analysis (least squares mean difference [LSMD] -2.347 , $p = 0.000$; LSMD -3.369 , $p = 0.000$; LSMD -3.454 , $p = 0.000$, respectively), as shown in Table 2 and Fig. 2.

Supporting the MMRM ANCOVA results, two-factor ANOVA showed that the difference between the two treatments was statistically significant, as indicated by the effect of group, using the between-subject factor [$F(1, 72) = 19.484$, $p = 0.000$, $\eta^2 = 0.213$]. The behavior of the two treatment approaches was not similar across time [group $\times$ time interaction, $F(3, 216) = 15.281$, $p = 0.000$, $\eta^2 = 0.175$].

Effect on Biological Markers

The differences in serum levels of TNF- α , IL-1 β , IL-6, BDNF, serotonin, IL-17, MDA, CRP, and vitamin B12 were not statistically significant between the placebo and MET groups at baseline ($p > 0.05$). The MET group showed a statistically significant decrease in the serum levels of TNF- α ,

Table 1 Demographic data of the participants

	Placebo group (n=40)	Metformin group (n=40)	Statistical value
Age (years)	35.1 ± 8.02	34.05 ± 8.4	$t = 0.572, df = 78, p = 0.569$
Gender			
Male	22 (55%)	23 (57.5%)	$\chi^2 = 0.051, df = 1, p = 0.822$
Female	18 (45%)	17 (42.5%)	$\chi^2 = 0.051, df = 1, p = 0.822$
Smoking	22 (55%)	23 (57.5%)	$\chi^2 = 0.051, df = 1, p = 0.822$
Weight (kg)	72.63 ± 4.67	70.88 ± 7.56	$t = 1.246, df = 78, p = 0.216$
Height (cm)	171.38 ± 6.96	170.68 ± 7.84	$t = 0.422, df = 78, p = 0.674$
BMI (kg/m ²)	24.74 ± 1.67	24.18 ± 1.57	$t = 1.822, df = 78, p = 0.072$
Marital status			
Single	7 (17.5%)	8 (20%)	$\chi^2 = 0.237, df = 1, p = 0.888$
Married	15 (37.5%)	13 (32.5%)	$\chi^2 = 0.237, df = 1, p = 0.888$
Divorced	18 (45%)	19 (47.5%)	$\chi^2 = 0.237, df = 1, p = 0.888$
HAM-D score	21 ± 1.29	21.24 ± 1.26	$t = 1.057, df = 78, p = 0.294$
Fasting blood glucose	93.33 ± 9.15	96.23 ± 9.2	$t = 1.956, df = 78, p = 0.166$
HbA1c	4.99 ± 0.7	4.73 ± 0.57	$t = 3.416, df = 78, p = 0.068$
Episodes of depression			
First	35 (87.5%)	34 (85%)	$\chi^2 = 0.105, df = 1, p = 0.745$
Second	5 (12.5%)	6 (15%)	$\chi^2 = 0.105, df = 1, p = 0.745$
Drugs used in last episode			
Fluoxetine	3 (7.5%)	3 (7.5%)	$\chi^2 = 0.158, df = 1, p = 0.924$
Sertraline	2 (5%)	3 (7.5%)	$\chi^2 = 0.157, df = 1, p = 0.692$

Data presented as mean ± SD. BMI, body mass index; HAM-D score, Hamilton Depression Rating Scale score; HbA1c, hemoglobin A1c

IL-1 β , IL-6, IGF-1, MDA, and CRP in comparison with the placebo group after 12 weeks of treatment as indicated by the effect of group and between-subject factor [$F(1, 71) = 2172.3, p = 0.000, \eta^2 = 0.970$; $F(1, 71) = 2281.8, p = 0.000, \eta^2 = 0.970$; $F(1, 71) = 37.09, p = 0.000, \eta^2 = 0.343$; $F(1, 71) =$

619.86, $p = 0.000, \eta^2 = 0.896$; $F(1, 71) = 2294.8, p = 0.000, \eta^2 = 0.970$; and $F(1, 71) = 135.96, p = 0.000, \eta^2 = 0.657$; $F(1, 71) = 4489.3, p = 0.000, \eta^2 = 0.984$, respectively].

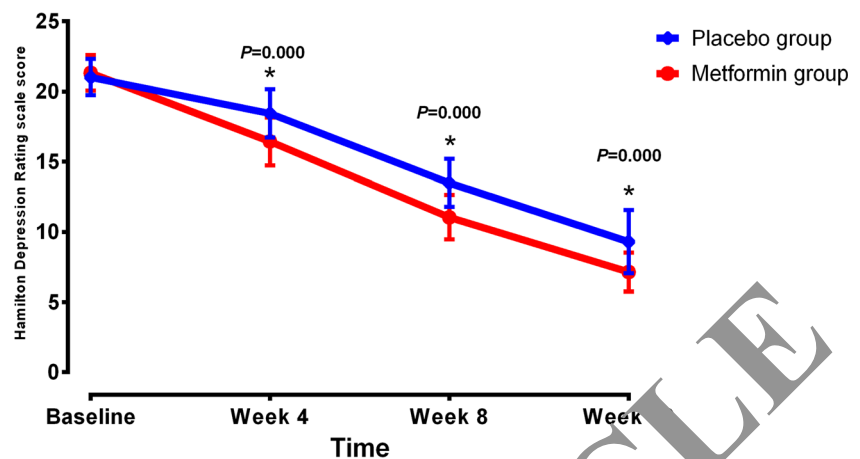
In contrast, the MET group exhibited a statistically significant increase in the serum levels of BDNF and serotonin

Table 2 Hamilton Depression Rating Scale score change from baseline to week 12

Outcome	Placebo group (n= 40)	Metformin group (n= 40)
MMRM		
Change at week 4, LSM (SE)	-2.525 (0.203)	-4.872 (0.227)
LSMD vs. placebo (95% CI)	---	-2.347 (-2.816 to -1.878)
<i>p</i> value	---	0.000
MMRM		
Change at week 8, LSM (SE)	-7.44 (0.417)	-10.809 (0.257)
LSMD vs. placebo (95% CI)	---	-3.369 (-4.043 to -2.695)
<i>p</i> value	---	0.000
MMRM		
Change at week 12, LSM (SE)	-11.428 (0.488)	-14.882 (0.309)
LSMD vs. placebo (95% CI)	---	-3.454 (-4.145 to -2.76)
<i>p</i> value	---	0.000

MMRM, mixed-effects model for repeated measures; SE, standard error; LSM, least squares mean; LSMD, least squares mean difference; CI, confidence interval

Fig. 2 Change in Hamilton Depression Rating Scale (HAM-D) total score from baseline to week 12. Data presented as mean and 95% confidence interval (CI)



compared with the placebo group [$F(1, 71) = 81.78, p = 0.000, \eta^2 = 0.535$, and $F(1, 71) = 54.54, p = 0.000, \eta^2 = 0.434$, respectively]. It is worth noting that TNF- α , IL-1 β , IL-6, IGF-1, MDA, and CRP serum levels showed a statistically significant decrease after 12 weeks of treatment relative to their baseline values in both groups, as indicated by the effect of group $\times$ time interaction [$F(1, 71) = 40.382, p = 0.000, \eta^2 = 0.353$; $F(1, 71) = 30.725, p = 0.000, \eta^2 = 0.302$; $F(1, 71) = 35.971, p = 0.000, \eta^2 = 0.336$; $F(1, 71) = 56.76, p = 0.000, \eta^2 = 0.444$; $F(1, 71) = 22.67, p = 0.000, \eta^2 = 0.242$; and $F(1, 71) = 16.758, p = 0.011, \eta^2 = 0.19$, respectively]. In contrast, a statistically significant increase was observed in the serum levels of BDNF and serotonin after 12 weeks of the treatment in the two groups compared with their baseline levels [$F(1, 71) = 28.78, p = 0.000, \eta^2 = 0.288$, and $F(1, 71) = 31.56, p = 0.000, \eta^2 = 0.307$, respectively], as shown in Table 3.

The difference in serum levels of vitamin B12 between baseline and after treatment was not statistically significant in either group [$F(1, 71) = 1.058, p = 0.307, \eta^2 = 0.014$ and $F(1, 71) = 2.056, p = 0.107, \eta^2 = 0.026$, respectively].

For further analysis of the data, the correlations between HAM-D score and each of the serum levels of TNF- α , IL-1 β , IL-6, IGF-1, MDA, CRP, BDNF, and serotonin were calculated for both groups at baseline and after treatment. The serum levels of TNF- α , IL-1 β , IL-6, IGF-1, MDA, and CRP were found to have a statistically significant positive correlation with HAM-D score before treatment ($r = 0.30, p = 0.025$; $r = 0.33, p = 0.009$; $r = 0.545, p = 0.031$; $r = 0.431, p = 0.002$; $r = 0.545, p = 0.001$; and $r = 0.652, p = 0.002$, respectively), and after treatment ($r = 0.774, p = 0.000$; $r = 0.719, p = 0.000$; $r = 0.676, p = 0.000$; $r = 0.701, p = 0.000$; $r = 0.576, p = 0.000$; and $r = 0.632, p = 0.000$, respectively).

In contrast, the serum levels of BDNF and serotonin showed a statistically significant negative correlation with HAM-D score before treatment ($r = -0.574, p = 0.021$; and $r = -0.548, p = 0.002$, respectively) and after treatment ($r = -0.694, p = 0.000$; and $r = -0.681, p = 0.001$, respectively).

Clinical Adverse Effects

The difference between the MET and placebo groups in the frequency of side effects was not statistically significant. Consequently, dropouts from therapy due to lack of efficacy or adverse events appeared to be limited. The most commonly reported adverse effects in both groups were nausea (13.5% placebo, 18.9% MET), vomiting (2.7% placebo, 5.4% MET), abdominal pain (8.1% placebo, 10.8% MET), heartburn (10.8% placebo, 13.5% MET), bloating (16.2% placebo, 18.9% MET), constipation (8.1% placebo, 10.8% MET), diarrhea (10.8% placebo, 8.1% MET), decreased appetite (10.8% placebo, 10.8% MET), increased appetite (13.5% placebo, 18.9% MET), fatigue (10.8% placebo, 13.5% MET), dry mouth (8.1% placebo, 10.8% MET), insomnia (16.2% placebo, 18.9% MET), headache (21.6% placebo, 18.9% MET), tremors (2.7% placebo, 5.4% MET), dizziness (10.8% placebo, 13.5% MET), sexual dysfunction (10.8% placebo, 13.5% MET), blurred vision (10.8% placebo, 13.5% MET), and sweating (10.8% placebo, 8.1% MET). The other reported adverse effects were transient and resolved spontaneously. Table 4 shows that the rate of adverse effects was not statistically different between the two groups.

Discussion

All previously published human studies on the role of MET in depression have been conducted in diabetic patients with concomitant MDD [24, 25]. Therefore, to the best of our knowledge, our study is the first adequately powered randomized, double-blind, placebo-controlled trial to evaluate the adjunctive role of MET in the management of MDD in adult patients without other comorbidities.

Despite the introduction of newer-generation antidepressants, approximately 50% of patients experience no response to treatment with first-line antidepressants [30]. Thus, it was reported that using a combination of

Table 3 Selected biological markers of the patients at baseline and after 12 weeks of treatment

Groups	Placebo group (n=40)			Metformin group (n=40)			* P value after 12 weeks
	Baseline	12 weeks	**P value	Baseline	12 weeks	**P value	
TNF- α (pg/mL)	10.22 $\pm$ 1.42	7.16 $\pm$ 0.99	$p = 0.000$	10.58 $\pm$ 1.28	5.27 $\pm$ 0.64	$p = 0.000$	$p = 0.000$
IL-1 β (pg/mL)	1.995 $\pm$ 0.25	0.75 $\pm$ 0.104	$p = 0.000$	1.93 $\pm$ 0.18	0.59 $\pm$ 0.109	$p = 0.000$	$p = 0.000$
IL-6 (pg/mL)	9.2 $\pm$ 1.28	6.46 $\pm$ 0.89	$p = 0.000$	9.52 $\pm$ 1.15	4.75 $\pm$ 0.58	$p = 0.000$	$p = 0.000$
IGF-1 (ng/mL)	210.82 $\pm$ 28.96	148.28 $\pm$ 20.43	$p = 0.000$	213.74 $\pm$ 21.9	109.13 $\pm$ 13.25	$p = 0.000$	$p = 0.000$
MDA (ng/mL)	4.6 $\pm$ 0.64	3.22 $\pm$ 0.44	$p = 0.000$	4.76 $\pm$ 0.58	2.37 $\pm$ 0.288	$p = 0.000$	$p = 0.000$
CRP (mg/L)	5.08 $\pm$ 0.71	3.3 $\pm$ 0.56	$p = 0.000$	5.26 $\pm$ 0.64	2.47 $\pm$ 0.46	$p = 0.000$	$p = 0.000$
BDNF (ng/mL)	18.7 $\pm$ 4.26	41.9 $\pm$ 6.64	$p = 0.000$	19.67 $\pm$ 5.3	59.77 $\pm$ 9.8	$p = 0.000$	$p = 0.000$
Serotonin (ng/mL)	74.83 $\pm$ 21.02	125.76 $\pm$ 19.91	$p = 0.000$	76.18 $\pm$ 18.54	161.64 $\pm$ 21.51	$p = 0.000$	$p = 0.000$
Vit B12 (pg/mL)	299.64 $\pm$ 41.75	301.72 $\pm$ 40.90	$p = 0.307$	295.06 $\pm$ 35.4	288.27 $\pm$ 34.66	$p = 0.107$	$p = 0.092$

Data presented as mean $\pm$ SD, Bonferroni adjusted (0.05 / 18)

TNF- α , tumor necrosis factor alpha; IL-1 β , interleukin-1 beta; IL-6, interleukin-6; IGF-1, insulin-like growth factor-1; MDA, malondialdehyde; CRP, C-reactive protein; BDNF, brain-derived neurotrophic factor; Vit B12, vitamin B12

*Between-group comparison after 12 weeks

**Within-group comparison

medications at the start of treatment may provide additional therapeutic benefits in MDD patients [31]. Regarding patient response to fluoxetine monotherapy, the response rate of 59% in our study is comparable to previously reported response rates of 50–59% for monotherapy in two studies conducted over 6 weeks [11, 32]. The remission rate of 46% in the fluoxetine monotherapy

group in our study is also comparable to the 5–45% remission rates in the above-mentioned studies [11, 32]. In addition, the response rate in our combination therapy group (89%) was comparable to the rate of 90% reported in previous studies. The longer duration of therapy of 12 weeks in our study compared with those previous trials may explain why the remission rate of 81% was higher

Table 4 Clinical complications and side effects reported as number per group

Side effects	Placebo group (n=40)	Metformin group (n=40)	Statistical value
Nausea	5	6	$\chi^2 = 0.105$, $df = 1$, $p = 0.745$
Vomiting	1	2	$\chi^2 = 0.346$, $df = 1$, $p = 0.556$
Abdominal pain	3	4	$\chi^2 = 0.157$, $df = 1$, $p = 0.692$
Heartburn	4	5	$\chi^2 = 0.125$, $df = 1$, $p = 0.723$
Bloating	6	7	$\chi^2 = 0.092$, $df = 1$, $p = 0.762$
Constipation	3	4	$\chi^2 = 0.157$, $df = 1$, $p = 0.692$
Diarrhea	4	3	$\chi^2 = 0.157$, $df = 1$, $p = 0.692$
Decreased appetite	6	4	$\chi^2 = 0.457$, $df = 1$, $p = 0.499$
Increased appetite	5	7	$\chi^2 = 0.392$, $df = 1$, $p = 0.531$
Fatigue	4	5	$\chi^2 = 0.125$, $df = 1$, $p = 0.723$
Dry mouth	3	4	$\chi^2 = 0.157$, $df = 1$, $p = 0.692$
Insomnia	6	7	$\chi^2 = 0.092$, $df = 1$, $p = 0.762$
Headache	8	7	$\chi^2 = 0.082$, $df = 1$, $p = 0.775$
Tremors	1	2	$\chi^2 = 0.346$, $df = 1$, $p = 0.556$
Dizziness	4	5	$\chi^2 = 0.125$, $df = 1$, $p = 0.723$
Sexual dysfunction	4	5	$\chi^2 = 0.125$, $df = 1$, $p = 0.723$
Blurred vision	4	5	$\chi^2 = 0.125$, $df = 1$, $p = 0.723$
Sweating	4	3	$\chi^2 = 0.157$, $df = 1$, $p = 0.692$

than that of previous studies (35–59%). Moreover, it has been reported that 31–41% of unimproved patients at the sixth week may experience remission at the 12th week [33]. In addition, the relatively high prevalence of patients with first-episode depression (86%) with lower severity based on HAM-D score in our study compared with the previous studies might explain the higher remission rate in our study [34, 35]. MET also caused a rapid reduction in the HAM-D score in the first 4 weeks, and the difference between the two groups remained highly significant until the end of the trial. These results are in line with previous studies which reported that anti-inflammatory agents may produce rapid onset of antidepressant effects in MDD patients [11, 36].

This clinical improvement in the MET group can be attributed to its anti-inflammatory, antioxidant, and neuroprotective effects, which led to the substantial reduction in the serum levels of TNF- α , IL-1 β , IL-6, IGF-1, MDA, and CRP together with a significant increase in the serum levels of BDNF and serotonin compared with their baseline values and with the placebo group [37, 38]. Our findings are consistent with other studies, which reported that MET decreased the expression of IL-1 β and IL-6 regardless of diabetes status [39, 40]. Moreover, MET decreases TNF- α -mediated gene expression of pro-inflammatory and cell adhesion molecules to inhibit endothelial cell inflammation [41, 42]. In fact, reduced levels of pro-inflammatory cytokines lead to increased bioavailability of serotonin through regulation of multiple metabolic pathways [43, 44].

MET affects brain plasticity by modulating the levels of neurotrophic factors including BDNF through activation of AMP-activated protein kinase (AMPK) and cAMP-response element binding protein (CREB), as reported in preclinical models [45]. More specifically, MET increases the expression of BDNF by enhancing CREB phosphorylation and promoting histone acetylation, while increasing the plasticity of the synaptic structure [45]. MET has also been reported to reduce IGF-1 levels, endogenously produced reactive oxygen species (ROS), and DNA damage [46].

BDNF is a key neurotrophin found to be involved in synaptic plasticity, and therefore plays a crucial role in depression [47, 48]. Several studies have shown that BDNF may mediate the therapeutic action of antidepressants [49, 50].

Regarding the placebo group, fluoxetine exerts an anti-inflammatory effect, which is mediated by the reduction in pro-inflammatory cytokines and the expression of free radicals [51, 52]. Fluoxetine can induce immunomodulatory effects through its impact on serotonergic neurons in the central nervous system [53]. These properties of fluoxetine are reflected in the significant decrease in the serum levels of TNF- α , IL-1 β , IL-6, IGF-1, MDA, and CRP, alongside a significant increase in the serum levels of BDNF and serotonin, relative to their baseline values.

Our results are in line with other studies reporting that fluoxetine can reduce IGF-1 serum levels [17] and increase the level of BDNF [54] in depressed patients. All of these findings show that fluoxetine alone is successful in reducing symptoms of depression in comparison with placebo, as reported in several studies [55, 56].

It is worth mentioning that no pharmacokinetic interactions between MET and fluoxetine were reported, as each is metabolized by different isoenzymes [57, 58]. Also, there were no clinically significant side effects, due to the short treatment period and small dosage of MET (1000 mg). The serum level of vitamin B12 was assessed, as it was reported that MET treatment may be associated with vitamin B12 deficiency in some patients [59]; however, levels were in the normal range in both groups before and after treatment.

The enhanced antidepressant effect in the combination therapy group can be attributed to the addition of MET. Therefore, our study showed that MET is an effective and safe adjunct to fluoxetine in patients with MDD, and provided substantial proof for the efficacy of MET in patients with MDD without additional comorbidities. This notion is strengthened in particular by earlier studies which suggested that MET could be used preferentially when considering the use of hypoglycemic agents for diabetic patients with depression-like symptoms [24, 25]. This is also in agreement with the results of preclinical studies indicating that MET produced antidepressant-like activity when given either alone or in combination with fluoxetine [23].

Recognizing biomarkers that are implicated in MDD pathophysiology is considered a clinical priority for physicians and psychiatrists in order to determine an appropriate treatment strategy [60]. Therefore, the serum levels of TNF- α , IL-1 β , IL-6, IGF-1, MDA, CRP, BDNF, and serotonin were evaluated and correlated with the HAM-D score to assess the biological effects of the trial medications. Serum levels of TNF- α , IL-1 β , IL-6, IGF-1, MDA, and CRP have been reported to be elevated in patients with MDD [8, 17, 52, 61]. These biomarkers were similarly elevated at baseline and decreased after treatment in our patients.

Several studies have reported that in drug-free major depressed subjects, the serum levels of BDNF and serotonin are lower than in normal controls [62, 63]. Accordingly, we found that the serum levels of BDNF and serotonin were lower at baseline and increased after intervention.

Nevertheless, this trial had some limitations, including a short follow-up period and the use of only a fixed dose of MET. In addition, our study lacks an assessment of MET metabolic activity in healthy MDD patients. Therefore, we recommend study replication with further investigation for a longer duration. In particular, it will be interesting to evaluate MET antidepressant efficacy without additional psychotropic drugs.

Conclusion

The antidiabetic MET improved the antidepressant effects, reflected clinically by better response and higher remission rates. Therefore, it represents a promising candidate for treating nondiabetic MDD patients. Moreover, detection of inflammatory markers and BDNF may be clinically useful in assessing antidepressant response. However, the limitations of the study would encourage researchers to conduct further investigations with a larger sample size and longer follow-up duration.

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Compliance with Ethical Standards

Conflict of Interest The authors declare that they have no conflict of interests.

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